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Electric Skateboard

Abstract

An electric skateboard comprises a main board body, a front axle assembly, a rear axle assembly, a buffering device, a moving device and an electric control device, wherein the electric control device is electrically connected with the front axle assembly and the rear axle assembly, the moving device supports and moves the skateboard, and the buffering mechanism is operatively connected to the main board body and the moving device, and provides buffering for the movement of the skateboard when the skateboard is in a moving state.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to the technical field of skateboards, in particular to an electric skateboard.

BACKGROUND

[0002] Skateboarding, as an extreme sport with a long history, originated in the United States in the 1950s. At first, it was just a land substitute for surfboard, aiming to make enthusiasts experience similar fun when they can't surf. However, with the passage of time, skateboarding has gradually developed into a unique and challenging sport, attracting the love of many young people around the world. Nowadays, skateboarders all over the world have combined skateboarding with other fields of art, culture, science and technology in their own way, constantly expanding the boundaries of this sport. As a new sub-field, the electric skateboard has many problems, such as motion instability at high speed, which is also called "death swing" by people in the industry. On the one hand, the body design of electric skateboard greatly affects its stability. Because the car body is light, it is easy to cause the risk of sideslip and rollover when driving at high speed. On the other hand, factors such as road conditions and wind direction during driving will also have an impact on the stability of the car body. In the complex road conditions and environment, the high-speed driving safety of electric skateboards is worrying. Therefore, how to ensure the stability and safety of electric skateboards is the direction that the industry needs to study deeply.

[0003] In the patent document with publication number of U.S. Pat. No. 20220331686A1, a skateboard suspension is introduced in detail, comprising a pair of steering knuckles on opposite sides of a chassis frame of the skateboard, a pair of upper control arms connected between the pair of steering knuckles and the chassis frame, a pair of lower control arms connected between the pair of steering knuckles and the chassis frame, and a first spring damper connected between the pair of upper control arms. This kind of skateboard has many parts such as swing arms and springs, which can ensure the comfort when driving. However, because the overall structure is too small and too thin, there are too many connection points between the parts, so there are problems such as short service life and insufficient stability.

[0004] Therefore, based on the above problems, we propose a brand-new electric skateboard structure, which has a stable and solid front axle assembly and rear axle assembly, and a stable and reliable buffering structure is also provided on the skateboard to ensure the stability of the driving process.

SUMMARY

[0005] The present invention provides an electric skateboard, which includes a main board body, a front axle assembly, a rear axle assembly, a moving device and an electric control device; [0006] wherein, the main board body has a front end and a rear end, the front axle assembly is arranged at the front end, the rear axle assembly is arranged at the rear end, the electric control device is electrically connected with the front axle assembly and the rear axle assembly, and the moving device supports and moves the skateboard; and [0007] wherein, the front axle assembly includes a front support component, and the front support component forms a rotatable connection with the main board body; and [0008] wherein, the rear axle assembly includes a driving device, a rear support component and at least one buffering mechanism, wherein the main board body is rotatably connected to the rear support component, and the buffering mechanism is operatively connected to the main board body and the moving device, and provides buffering for a movement of the skateboard when the skateboard is in a moving state.

[0009] The present invention also provides an electric skateboard, which includes a main board body, a front axle assembly, a rear axle assembly, a moving device and an electric control device;

[0010] wherein, the main board body has a front end and a rear end, the front axle assembly is arranged at the front end, the rear axle assembly is arranged at the rear end, the electric control device is electrically connected with the front axle assembly and the rear axle assembly, and the

moving device supports and moves the skateboard; and [0011] wherein, the front axle assembly includes a front support component and at least one buffering mechanism, wherein the front support component is rotatably connected with the main board body, and the buffering mechanism is operatively connected to the main board body and the mobile device, and provides buffering for a movement of the skateboard when the skateboard is in a moving state; and [0012] wherein, the rear axle assembly includes a driving device and a rear support component, and the main board body is rotatably connected to the rear support component.

[0013] The present invention also provides a buffering mechanism of an electric skateboard, which includes a support structure, a damper assembly and a restorer assembly, wherein the support structure includes at least one fixed support component and a movable support component; the fixed support component is fixedly connected with the damper assembly and the restorer assembly; the damper assembly is provided with a damper push rod and a damper shell which can slide on the damper push rod; and the restorer assembly includes a restorer fixing assembly and at least one elastic member; two ends of the elastic member are pressed against two ends of the restorer fixing component, the movable support component is operatively connected with the damper assembly and the restorer assembly, and the damper assembly further includes at least one damping adjusting device which is arranged on the damper shell and can change a damping size of the damper assembly.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0014] In order to explain the technical scheme of this application more clearly, the drawings needed in the implementation will be briefly introduced below. Obviously, the drawings described below are only some implementations of this application. For those skilled in the art, other drawings can be obtained according to these drawings without creative work.

[0015] FIG. 1 is a schematic perspective view of a skateboard;

[0016] FIG. 2 is a schematic diagram of the overall explosion of the skateboard;

[0017] FIG. 3 is a schematic perspective view of the electronic control device;

[0018] FIG. 4 is a schematic perspective view of the control assembly;

[0019] FIG. 5 is a perspective view of the front axle assembly;

[0020] FIG. 6 is an explosion schematic diagram of the front axle assembly;

[0021] FIG. 7 is a schematic diagram of the rear explosion of the front axle assembly;

[0022] FIG. 8 is a schematic perspective view of the bridge arm;

[0023] FIG. 9 is an explosion schematic diagram of the buffering structure;

[0024] FIG. 10 is a schematic perspective view of the bridge support;

[0025] FIG. 11 is an explosion schematic diagram of the restorer assembly;

[0026] FIG. 12 is a schematic perspective view of the skateboard viewed from the rear;

[0027] FIG. 13 is a schematic perspective view of the rear axle assembly;

[0028] FIG. 14 is an explosion schematic diagram of the driving device;

[0029] FIG. 15 is a schematic perspective view of the driving device for removing the dust cover;

[0030] FIG. 16 is a schematic perspective view of the driving device viewed from the rear.

[0031] In the figures:

[0032] Main board body (100); Board surface (101); Antiskid member (102); Headlamp (103); Tail lamp (104); Front end (106); Rear end (107); Tire (110); Wheel hub (112); Electric control device (200); Sealing gasket (201); Battery armor (203); Battery compartment cover (205); Battery core (206); Battery circuit board (207); Battery inlet circuit (209); Battery discharge circuit (211); First light connection port (213); First light cable (215); BMS (216); Controller shell (240); First control circuit board (250); First control circuit component (251); Light control circuit board (260); Light

control circuit component (261); Antenna assembly (265); Power switch (267); Charging port (269); First motor cable (271); Second motor cable (272); Tail lamp cable (273); Cable connector (274); Motor cable clamp (276); Front axle assembly (300); Front support component (305); Fixed support component (310); Movable support component (315); Front axle (316); Bridge arm (317); Damping push rod installation cavity (318); Support shaft installation cavity (319); Bridge support (320); Central shaft cavity (321); Support shaft hole (322); Central shaft hole (323); Damping hole (324); Support rotating shaft (325); Bridge damping block (326); Damper assembly (330); Damping adjusting device (331); Damper lower protective plate (333); Damper shell (335); Damper push rod (337); Damper slider (339); Damper clamp (340); Sliding cavity (341); Restorer assembly (360); Restorer fixing seat (362); Movable shaft block (364); Restorer adjusting nut (366); Central shaft sleeve (368); Central shaft connector (369); Restorer spring (370); Spring first base (372); First connecting shaft (373); Restoring rubber (374); Spring second base (375); Rear axle assembly (400); Driving device (410); Reinforcing rod (412); Rear axle (414); Motor (416); Driving substrate (418); Driving gear (420); Driven gear (422); Transmission belt (424); Dust cover (426).

DESCRIPTION OF EMBODIMENTS

[0033] In the following, the technical scheme in the embodiment of the application will be clearly and completely described with reference to the drawings in the embodiment of the application. Obviously, the described embodiment is only a part of the embodiment of the application, but not the whole embodiment. Based on the embodiments in this application, all other embodiments obtained by those skilled in the art without creative labor belong to the protection scope of this application.

[0034] Reference to “an example” or “an embodiment” herein means that a particular feature, structure or characteristic described in connection with an embodiment or an embodiment can be included in at least one embodiment of this application. The appearance of this phrase in various places in the specification does not necessarily refer to the same embodiment, nor is it an independent or alternative embodiment mutually exclusive with other embodiments. It is understood explicitly and implicitly by those skilled in the art that the embodiments described herein can be combined with other embodiments.

[0035] In this specification, for the sake of convenience, words and expressions indicating orientation or positional relationship such as “middle”, “upper”, “lower”, “front”, “rear”, “vertical”, “horizontal”, “top”, “inner” and “outer” are used to illustrate the positional relationship of constituent elements with reference to the attached drawings, only for the convenience of description. The positional relationship of the constituent elements is appropriately changed according to the direction of the described constituent elements. Therefore, it is not limited to the words and expressions described in the specification, and can be replaced appropriately according to the situation.

[0036] As shown in FIGS. 1 and 2, the present invention provides an electric skateboard, which includes a main board body 100, a front axle assembly 300 and a rear axle assembly 400, wherein the main board body 100 has a front end 106 and a rear end 107, and the front axle assembly 300 is arranged at the front end 106 and the rear axle assembly 400 is arranged at the rear end 107.

[0037] The skateboard also includes an electric control device 200 and a moving device arranged below the main board body 100, wherein the electric control device 200 is electrically connected with the front axle assembly 300 and the rear axle assembly 400, and the moving device supports and moves the skateboard.

[0038] The moving device can be a wheel, a mechanical foot, a crawler, etc. In this embodiment, the moving device includes a wheel hub 112 and a tire 110 fixed on the wheel hub 112. As the core component of the mobile device, the wheel hub 112 bears the heavy responsibility of bearing the tire 110 and the whole equipment, and the tire 110 is responsible for providing good road holding and stability during contact with the ground. The wheel hub 112 can be made of aluminum alloy,

steel and other materials to meet different scenes and load requirements, but it is not limited thereto and can be any other materials that can meet the requirements. Tire **110** is the part where the mobile device contacts the ground, and its performance directly affects the running performance and stability of the equipment. The material of the tire **110** needs to have high strength and wear resistance to ensure a long service life. According to different terrain and load conditions, the tire **110** can adopt different tread patterns to provide the tire **110** with different performances such as good road holding, wear resistance and skid resistance. In practical application, according to different scenes and requirements, a suitable combination of wheel hub **112** and tire **110** can be selected to achieve efficient and stable mobility.

[0039] The skateboard also includes a headlamp **103**, which is arranged at the front end **106** of the main board body **100** to improve the comfort and safety of users. The headlamp **103** adopts a LED light source, which has the advantages of high brightness, low energy consumption and long life.

[0040] In other embodiments, the headlamp **103** may also be a fluorescent lamp, a gas discharge lamp, or the like.

[0041] In another embodiment, the headlamp **103** is also equipped with an intelligent function, which can automatically adjust the brightness according to the ambient light, further improving the driving safety.

[0042] The skateboard further includes an electric control device **200** arranged at the bottom of the main board body **100**, and a first light cable **215** is arranged at one end of the electric control device **200** near the front end **106**, and the headlamp **103** is electrically connected with the first light cable **215**.

[0043] The rear axle assembly **400** is provided with a driving device **410**. In this embodiment, the driving device **410** is mainly driven by a motor.

[0044] In another embodiment, the driving device **410** can also be driven by a fuel engine, hybrid power or other new energy sources.

[0045] A motor cable is arranged at one end of the electronic control device **200** near the rear end of the main board body **100**, and the motor cable connects the electronic control device **200** and the driving device **410**. The motor cable includes a first motor cable **271** and a second motor cable **272**, one end of the first motor cable **271** is connected to the electronic control device **200**, and one end of the second motor cable **272** is connected to the driving device **410**. The first motor cable **271** and the second motor cable **272** are electrically connected through a cable connector **274**.

[0046] The main board body **100** includes a board surface **101** and at least one anti-skid member **102** arranged on the board surface **101**. In this embodiment, the anti-skid member **102** is anti-skid sandpaper, which has a rough surface and can increase the friction force of a user stepping on a skateboard. The anti-skid member **102** is composed of a plurality of anti-skid sandpapers, which not only enhances the personality of the skateboard, but also avoids the problem that the anti-skid member **102** needs to be replaced as a whole after being damaged, and the damaged anti-skid sandpaper can be replaced separately.

[0047] In another embodiment, the anti-slip member **102** may be made of a rubber material. Rubber has good elasticity and wear resistance, which can effectively increase the friction on the surface of skateboard and play a role in anti-skid.

[0048] In another embodiment, some special coating materials can also be used to make the anti-skid member **102**. They can increase the friction and improve the anti-skid performance by forming an anti-skid coating on the surface. These alternative materials can provide anti-skid effect similar to sandpaper, but also have their own unique characteristics and applicable scenarios. Therefore, the choice of anti-skid materials should be comprehensively considered according to the actual needs and conditions.

[0049] In another embodiment, the front axle assembly **300** is also provided with at least one anti-collision bar. When a vehicle collides, the anti-collision bar can effectively absorb the impact force and reduce the impact on the car body and other components. When a vehicle collides, the bumper

often bears the impact force first, thus protecting other parts of the vehicle from damage. In this way, the maintenance cost of the skateboard has been effectively controlled.

[0050] In this embodiment, the skateboard is controlled by a remote controller. In another embodiment, the skateboard can also be controlled by somatosensory induction, resistance induction and mechanical buttons on the board.

[0051] In another embodiment, the tire **110** is also provided with a fender, which can improve the comfort of using on muddy road sections.

[0052] In another embodiment, the main board body **100** is also provided with a detachable pull rod to improve the operability of the skateboard and lower the operating threshold.

[0053] As shown in FIGS. 2 and 3, the skateboard also includes an electric control device **200** arranged at the bottom of the main board body **100**, which includes a battery assembly and a control assembly. The battery assembly includes a sealing gasket **201** and a battery compartment cover **205** fixed at the bottom of the main board body **100**. The battery compartment cover **205** is covered with at least one battery armor **203**, which can protect the battery compartment cover **205**. In this example, there are three battery armors **203**. The material and structural form of battery armor **203** can be selected and designed according to different application scenarios and requirements, including the use of metals, ceramics, composite armor, etc.

[0054] In another embodiment, the battery armor **203** is made of lightweight materials such as aluminum alloy and titanium alloy, which can provide good impact resistance without increasing the load.

[0055] In another embodiment, the battery armor **203** is made of a composite armor, which is composed of multiple materials. Through the combination and optimization of different materials, the protection ability can be improved.

[0056] A battery circuit board **207** and a battery core **206** are arranged in the battery compartment cover **205**, the battery core **206** is electrically connected with the battery circuit board **207**, and the battery core **206** is fixed on the battery circuit board **207** through a battery bracket, and a BMS (battery management system) is arranged on the battery circuit board **207**, and the BMS **216** is electrically connected with the battery circuit board **207**. BMS **216** is responsible for monitoring the working parameters of battery core **206** in real time to ensure that the battery operates in the best working condition.

[0057] BMS **216** is arranged on the battery circuit board **207**, which further improves the

[0058] safety of the battery. BMS **216** can monitor the working state of battery core **206** in real time, and take immediate measures to protect it when abnormal conditions are found. For example, when the temperature of the battery core **206** is too high, the BMS **216** will start the cooling system to reduce the temperature of the battery core **206** and prevent the battery core **206** from being damaged by overheating. In addition, BMS **216** can effectively prevent overcharge and overdischarge of battery core **206** and ensure the safety of battery core **206** during use.

[0059] The battery circuit board **207** is provided with a battery inlet circuit **209** and a battery discharge circuit **211**. The battery core **206** is electrically connected with the battery circuit board **207**, the battery core **206** is electrically connected with the battery inlet circuit **209** and the battery discharge circuit **211**, and one end of the battery compartment cover **205** near the front end **106** is provided with a first light connection port **213** which is connected with a first light cable **215**, and the battery core **206** is connected with the first light cable **215** and the first light connection port **213**.

[0060] In this embodiment, the battery inlet circuit **209** adopts the 18AWG specification, and the battery discharge circuit **211** adopts the 13AWG specification, but it is not limited thereto, and it can also be any other specifications that meet the requirements.

[0061] As shown in FIGS. 3 and 4, the control assembly includes a controller shell **240**, a light control circuit component and a first control circuit component, wherein the light control circuit component and the first control circuit component are arranged in the controller shell **240**; the

battery core **206** is electrically connected with the light control circuit component and the first control circuit component; one end of the battery compartment cover **205** far from the front end **106** is connected with the control assembly; and the battery inlet circuit **209** and the battery inlet circuit **209** are respectively electrically connected with the control assembly.

[0062] Wherein, the first control circuit component includes a first control circuit board **250** and a first control circuit component **251** disposed on the first control circuit board **250**. The first control circuit component **251** includes a motor controller, a driver, a sensor, an encoder, and the like. These components are responsible for receiving and processing signals from the system, and controlling parameters such as starting, stopping, direction and speed of the motor according to instructions, so as to realize accurate movement and position control of the motor.

[0063] Wherein, the light control circuit component includes a light control circuit board **260** and a light control circuit component **261** arranged on the light control circuit board **260**. The light control circuit component **261** includes a light controller, a sensor, a dimmer, a relay and the like. These components are responsible for receiving and processing signals from the system, and controlling the switch, brightness, color temperature and other parameters of the light according to instructions to meet the needs of users.

[0064] One end of the control assembly near the rear end **107** of the main board body **100** is provided with an antenna assembly **265**, a power switch **267**, a charging port **269**, a motor cable and a tail lamp cable **273**.

[0065] The antenna assembly **265** is electrically connected with the control assembly, the power switch **267** is electrically connected with the control assembly, the charging port **269** is electrically connected with the control assembly, the motor cable is electrically connected with the control assembly, and the tail lamp cable **273** is electrically connected with the control assembly.

[0066] In another embodiment, the control assembly also includes an adjustable speed limit system to avoid the danger caused by the skateboard's speed increasing too fast when passing through the downhill section.

[0067] As shown in FIGS. **1** and **5**, the front axle assembly **300** includes a front support component **305** and at least one buffering mechanism, which includes a support structure and a front axle **316**, and the buffering mechanism includes a damper assembly **330** and a restorer assembly **360**. The damper assembly **330** also includes at least one damper adjusting device **331** arranged on a damper shell **335**, and the damper adjusting device **331** can change the damping of the damper assembly **330**.

[0068] In this embodiment, there are two buffering devices, which are respectively arranged on the front axle assembly **300** and the rear axle assembly **400**. The front axle assembly **300** and the rear axle assembly **400** are rotatably connected to the main board body **100**. When the skateboard is in motion, the buffering mechanism provides buffering for the movement of the skateboard through the front axle assembly **300** and the rear axle assembly **400**.

[0069] In another embodiment, only one buffering device can be provided, only on the front axle assembly **300**. The front axle assembly **300** and the rear axle assembly **400** are rotatably connected to the main board body **100**. When the skateboard is in motion, the buffering mechanism provides buffering for the movement of the skateboard through the front axle assembly **300**.

[0070] In another embodiment, only one buffering device can be provided, only on the rear axle assembly **400**. The front axle assembly **300** and the rear axle assembly **400** are rotatably connected to the main board body **100**. When the skateboard is in motion, the buffering mechanism provides buffering for the movement of the skateboard through the rear axle assembly **400**.

[0071] In another embodiment, only one buffering device can be provided, and the buffering device is simultaneously connected with the front axle assembly **300** and the rear axle assembly **400**, the front axle assembly **300** and the rear axle assembly **400** are rotatably connected to the main board body **100**. When the skateboard is in motion, the buffering mechanism provides buffering for the movement of the skateboard through the front axle assembly **300** and the rear axle assembly **400**.

[0072] As shown in FIG. 1, FIG. 5 to FIG. 7, the support structure includes a fixed support component **310** and a movable support component **315**. The fixed support component **310** fixedly connects the damper assembly **330** and the restorer assembly **360**, and the damper assembly **330** has a damper push rod **337** and a damper shell **335** that can slide on the damper push rod **337**. [0073] The restorer assembly **360** includes a restorer fixing component and at least one elastic member, two ends of which are pressed against two ends of the restorer fixing component, and the movable support component **315** operatively connects the damper assembly **330** and the restorer assembly **360**.

[0074] The front axle **316** is arranged on the bridge arm **317**, and the wheel hub **112** is rotatably connected with the front axle **316**.

[0075] As shown in FIGS. 6 and 8, the fixed support component **310** includes a bridge arm **317**, which has a damping push rod installation cavity **318** and a support shaft installation cavity **319**. The damper push rod **337** is mounted in the damping push rod installation cavity **318**, and a damper clamp **340** is arranged on the damper shell **335**. The damper clamp **340** has a sliding cavity **341**, and a damper slider **339** is slidably arranged in the sliding cavity **341**, and the damper slider **339** is rotatably connected in the damping hole **324**. A damper lower protective plate **333** is arranged below the damper push rod **337**, and the damper lower protective plate **333** can prevent damage caused by objects such as stones on the ground. As shown in FIG. 6, FIG. 8 to FIG. 10, the movable support component **315**

[0076] includes a bridge support **320** having a central shaft cavity **321**, a support shaft hole **322**, a central shaft hole **323** and a damping hole **324**. A support rotating shaft **325** is rotatably arranged in the support shaft installation cavity **319**, and the support rotating shaft **325** is rotatably connected with the support shaft hole **322**. At least one bridge damping block **326** is fixed on the bridge support **320**. When the bridge support **320** reaches the maximum swing amplitude, the bridge damping block **326** will contact the bridge arm **317**, and the bridge damping block **326** can slow down the impact during movement, ensuring safety and comfort.

[0077] In this embodiment, the bridge damping block **326** is made of a rubber material, which has good elasticity and damping performance, low cost, easy processing and convenient use.

[0078] In another embodiment, the bridge damping block **326** adopts a steel spring, and the steel spring damping block has the advantages of good damping effect, good durability and good fatigue resistance.

[0079] In another embodiment, the bridge damping block **326** is made of a viscoelastic material: viscoelastic material is a polymer material with good elasticity and damping performance, good damping effect and good durability.

[0080] In another embodiment, the composite material used in the bridge damping block **326** is a material composed of two or more materials, which can be combined according to different requirements to achieve the best damping effect.

[0081] To sum up, the bridge damping block **326** can be made of rubber, composite material, steel spring and viscoelastic material. In practical application, appropriate materials and structures should be selected according to specific conditions to achieve the best vibration reduction effect.

[0082] As shown in FIG. 6 and FIG. 11, the reducer assembly **360** includes at least one movable central shaft block **364** with a central shaft sleeve **368**, and a central shaft connector **369** rotatably connected with the central shaft hole **323** is arranged in the central shaft sleeve **368**, and the central shaft connector **369** is connected with the movable central shaft block **364** and the bridge support **320**, and the movable central shaft block **364** is arranged in the central shaft cavity. In this embodiment, the central shaft sleeve **368** is made of graphite. Graphite material has good wear resistance and vibration resistance, and the manufacturing process is relatively simple, which can reduce the production cost.

[0083] In another embodiment, the central shaft sleeve **368** may be made of ceramics. Ceramic materials have high hardness, high strength and excellent wear resistance, so ceramic bushings

show good performance in many mechanical equipment with high temperature, high load and corrosive environment.

[0084] In another embodiment, the central shaft sleeve **368** may be made of a metal matrix composite material. This kind of material combines the high strength of metal and the high wear resistance of composite materials, and has excellent properties.

[0085] The restorer fixing component includes a spring first base **372** and a spring second base **375** which are oppositely arranged; the elastic member includes a restorer spring **370** and a restoring rubber **374** which are arranged between the spring first base **372** and the spring second base **375**; both ends of the spring second base **375** are respectively connected with the restoring rubber **374** and the movable axle block **364**; and the restorer fixing component also includes a restorer fixing seat arranged on the bridge arm **317**.

[0086] The restorer assembly **360** also includes a first connecting shaft **373** arranged in the first spring base **372**. One end of the first connecting shaft **373** far from the first spring base **372** is arranged in the movable central shaft block **364**, and a restorer adjusting nut **366** is operatively arranged on the first connecting shaft **373**, which can change the elastic force of the elastic member.

[0087] As shown in FIG. 1, FIG. 6 to FIG. 11, when the buffering mechanism is in a normal state, the support shaft hole **322**, the central shaft hole **323** and the damping hole **324** of the bridge bearing **320** are approximately perpendicular to the damper assembly **330**, the elastic forces of the two restorer springs **370** are approximately equal, and the position of the damper shell **335** on the damper push rod **337** is approximately centered. When the skateboard turns or swings, the main board body **100** will tilt to either side. The restorer spring **370** of the restorer will be subjected to reaction force, and the restorer spring **370** in the opposite direction inclined to the main board body **100** will be compressed. At the same time, the bridge support **320** pushes the damper shell **335** to move in the opposite direction inclined to the main board body **100** through the damper slider **339** to generate damping force in the opposite direction, and the force generated when the main board body **100** is inclined is balanced by the reaction force and damping force of the spring, thus ensuring the stability during movement.

[0088] As shown in FIGS. 12 to 16, the rear axle assembly **400** includes a driving device **410**, a rear support component and at least one buffering mechanism. The rear support component includes a support structure, a rear axle **414** and a reinforcing rod **412**. The buffering mechanism includes a damper assembly **330** and a restorer assembly **360**. The reinforcing rod **412** helps to stabilize the driving device **410**, and can reduce the damage of the motor **416** to some extent.

[0089] Wherein, the main board body **100** is rotatably connected to the rear support component, and the buffering mechanism is operatively connected to the main board body **100** and the mobile device, and provides buffering for the movement of the skateboard when the skateboard is in a moving state.

[0090] The skateboard also has a tai lamp **104**, which is arranged at the rear end of the main board body **100**, and is electrically connected with a tai lamp cable **273**.

[0091] As shown in FIGS. 14 to 16, the driving device **410** includes a driving substrate **418** and a motor **416** fixed on the driving substrate **418**. The motor **416** has an output end, and a driving gear **420** is provided on the output end.

[0092] Wherein, the driving device **410** further includes a driven gear **422**, a transmission belt **424** and a dust cover **426**, wherein the driven gear **422** is rotatably connected to the driving substrate **418**; the transmission belt **424** movably connects the driven gear **422** and the driving gear **420**; and the dust cover **426** is installed on the driving substrate **418** to cover the driving gear **420**. The dust cover **426** can protect the driving gear **420** and prevent the driving gear **420** from being damaged by external dust, gravel, hair and other fine objects.

[0093] Wherein, the driven gear **422** is provided with a wheel fixing groove, and the wheel hub **112** is provided with a wheel fixing plate matched with the wheel fixing groove, and the driven gear

422 drives the wheel hub **112**.

[0094] Wherein, the driving device **410** further includes a motor cable clamp **276**, which is fixed on the bridge support **320**, and one end of the second motor cable **272** is connected to the cable connector **274**, and the motor cable clamp **276** fixes the cable connector **274**. In this embodiment, the skateboard has two motors **416** and two driving substrates

[0095] **418**, which are respectively arranged at the left and right ends of the rear axle assembly **400**. This design enables the skateboard to obtain stronger power output and more stable driving performance during driving. With the motors **416** and the driving base plates **418** distributed on the left and right sides, the skateboard can achieve an efficient and balanced driving effect and adapt to various complicated road conditions.

[0096] In another embodiment, the skateboard has a motor **416**, which is arranged in the middle of the rear axle assembly **400** and has two output shafts. This motor **416** is cleverly arranged in the middle of the rear axle assembly **400**, which can ensure that the center of gravity of the motor **416** coincides with that of the skateboard, thus improving the driving stability.

[0097] In another embodiment, the skateboard has a motor **416** with only one output end, but the single output end of the motor **416** is converted into two independent power output channels through clever gear design to output power for the wheel hubs **112** on both sides. In this way, the skateboard can still maintain good dynamic performance and stability during driving.

[0098] To sum up, through the different designs of the motor **416** of the skateboard, the efficient, stable and low-cost driving solutions in different needs and scenarios have been achieved. These embodiments show the diversity and flexibility of the skateboard drive system design, and provide more choices for consumers. At the same time, we will continue to study more innovative design of skateboard drive system to meet market demand and enhance user experience.

[0099] The principles and modes of operation of the present invention have been described in various embodiments thereof. It should be noted, however, that the present invention may be practiced otherwise than as specifically shown and described without departing from its scope.

Claims

1. An electric skateboard, comprising a main board body, a front axle assembly, a rear axle assembly, a moving device and an electric control device; central wherein, said main board body has a front end and a rear end, said front axle assembly is arranged at said front end, said rear axle assembly is arranged at said rear end, said electric control device is electrically connected with said front axle assembly and said rear axle assembly, and said moving device supports and moves said skateboard; and wherein, said front axle assembly comprises a front support component, and said front support component forms a rotatable connection with said main board body; and wherein, said rear axle assembly comprises a driving device, a rear support component and at least one buffering mechanism, wherein said main board body is rotatably connected to said rear support component, and said buffering mechanism is operatively connected to said main board body and said moving device, and provides buffering for a movement of said skateboard when said skateboard is in a moving state.

2. The electric skateboard according to claim 1, wherein said electric control device is arranged at a bottom of said main board body, and comprises a battery assembly, wherein said battery assembly comprises a sealing gasket and a battery compartment cover fixed at the bottom of said main board body; said battery compartment cover is covered with at least one battery armor, and a battery circuit board and a battery core are arranged in said battery compartment cover; and wherein, said battery circuit board is provided with a battery inlet circuit and a battery discharge circuit, and said battery core is electrically connected with said battery circuit board.

3. The electric skateboard according to claim 1, wherein said rear support component comprises a support structure, a rear axle and a reinforcing rod, and said buffering mechanism comprises a

damper assembly and a restorer assembly; and wherein, said support structure comprises at least one fixed support component and a movable support component, said fixed support component is fixedly connected with said damper assembly and said restorer assembly, said damper assembly is provided with a damper push rod and a damper shell which can slide on said damper push rod; said restorer assembly comprises a restorer fixing component and at least one elastic member, and two ends of said elastic member are pressed against two ends of said restorer fixing component; said movable support component is operatively connected with said damper assembly and said restorer assembly; and said damper assembly further comprises at least one damping adjusting device, which is arranged on said damper shell and can change a damping size of said damper assembly.

4. The electric skateboard according to claim 3, wherein said movable support component comprises a bridge support, and said bridge support is provided with a central shaft cavity, a support shaft hole, a central shaft hole and a damping hole; and wherein, said fixed support component comprises a bridge arm, said bridge arm is provided with a damping push rod installation cavity and a support shaft installation cavity, said damper push rod is installed in said damping push rod installation cavity, a damper clamp is arranged on said damper shell, at least one sliding cavity is arranged on said damper clamp, at least one damper slider is arranged in said sliding cavity in a sliding way up and down, and said damper slider is rotatably connected in said damping hole; and wherein, a support rotating shaft is rotatably arranged in said support shaft installation cavity, and said support rotating shaft is rotatably connected with said support shaft hole; and wherein at least one bridge damping block is fixed on said bridge support.

5. The electric skateboard according to claim 3, wherein said restorer assembly comprises at least one movable central shaft block, and said movable central shaft block is provided with a central shaft sleeve, and a central shaft connector is rotatably arranged in said central shaft sleeve, and said central shaft connector is rotatably connected with said movable support component; and said restorer fixing component comprises a spring first base and a spring second base which are oppositely arranged; said elastic member comprises a restorer spring and a restoring rubber arranged between said spring first base and said spring second base; two ends of said spring second base are respectively connected with said restoring rubber and said movable central shaft block; said restorer fixing component further comprises a restorer fixing seat arranged on said fixed support component; and said restorer fixing seat is connected with said spring first base.

6. The electric skateboard according to claim 5, wherein said restorer assembly further comprises a first connecting shaft arranged in said spring first base, and one end of said first connecting shaft far from said spring first base is arranged in said movable central shaft block, and a restorer adjusting nut is operatively arranged on said first connecting shaft, and said restorer adjusting nut can change an elastic force of said elastic member.

7. The electric skateboard according to claim 1, wherein said moving device comprises a wheel hub and a tire fixed on said wheel hub; and said driving device comprises a driving substrate and a motor fixed on said driving substrate, wherein said motor has at least one output end, and a driving gear is arranged on said output end; and wherein, said driving device further comprises a driven gear, a transmission belt and a dust cover, wherein said driven gear is rotatably connected to said driving substrate, said transmission belt is movably connected with said driven gear and said driving gear, and said dust cover is installed on said driving substrate to cover said driving gear; and wherein said driven gear is provided with a wheel fixing groove, said wheel hub is provided with a wheel fixing plate matched with said wheel fixing groove, and said driven gear drives said wheel hub; and wherein, said driving device further comprises a motor cable clamp, and said motor cable clamp is fixed on said rear axle assembly.

8. The electric skateboard according to claim 1, wherein said electric control device comprises a control assembly, which comprises a controller shell, a light control circuit component and a first control circuit component, wherein said light control circuit component and said first control circuit component are arranged in said controller shell; said battery core is electrically connected with said

light control circuit component and said first control circuit component; one end of said battery compartment cover far from said front end is connected with said control assembly; and said battery inlet circuit and said battery discharge circuit are respectively electrically connected with said control assembly.

9. An electric skateboard, comprising a main board body, a front axle assembly, a rear axle assembly, a moving device and an electric control device; wherein, said main board body has a front end and a rear end, said front axle assembly is arranged at said front end, said rear axle assembly is arranged at said rear end, said electric control device is electrically connected with said front axle assembly and said rear axle assembly, and said moving device supports and moves said skateboard; and wherein, said front axle assembly comprises a front support component and at least one buffering mechanism, wherein said front support component is rotatably connected with said main board body, and said buffering mechanism is operatively connected to said main board body and said mobile device, and provides buffering for a movement of said skateboard when said skateboard is in a moving state; and wherein, said rear axle assembly comprises a driving device and a rear support component, and said main board body is rotatably connected to said rear support component.

10. The electric skateboard according to claim 9, wherein said electric control device is arranged at a bottom of said main board body, and comprises a battery assembly, wherein said battery assembly is fixed on a sealing gasket and a battery compartment cover at the bottom of said main board body; said battery compartment cover is covered with at least one battery armor, and a battery circuit board and a battery core are arranged in said battery compartment cover; and wherein, said battery circuit board is provided with a battery inlet circuit and a battery discharge circuit, and said battery core is electrically connected with said battery circuit board.

11. The electric skateboard according to claim 9, wherein said front support component comprises a support structure and a front axle, and said buffering mechanism comprises a damper assembly and a restorer assembly; and wherein, said support structure comprises at least one fixed support component and a movable support component, said fixed support component is fixedly connected with said damper assembly and said restorer assembly, said damper assembly is provided with a damper push rod and a damper shell which can slide on said damper push rod; said restorer assembly comprises a restorer fixing component and at least one elastic member, and two ends of said elastic member are pressed against two ends of said restorer fixing component; said movable support component is operatively connected with said damper assembly and said restorer assembly; said damper assembly also comprises at least one damping adjusting device, which is arranged on said damper shell and can change a damping size of said damper assembly.

12. The electric skateboard according to claim 11, wherein said movable support component comprises a bridge support, and said bridge support is provided with a center shaft cavity, a support shaft hole, a center shaft hole and a damping hole; and wherein, said fixed support component comprises a bridge arm, and said bridge arm is provided with a damping push rod installation cavity and a support shaft installation cavity; said damper push rod is installed in said damping push rod installation cavity, and a damper clamp is arranged on said damper shell; at least one sliding cavity is arranged on said damper clamp, a damper slider is arranged in said sliding cavity in a sliding way up and down, and said damper slider is rotatably connected in said damping hole; and wherein, a support rotating shaft is rotatably arranged in said support shaft installation cavity, and said support rotating shaft is rotatably connected with said support shaft hole; and wherein at least one bridge damping block is fixed on said bridge support.

13. The electric skateboard according to claim 11, wherein said restorer assembly comprises at least one movable central shaft block, and said movable central shaft block is provided with a central shaft sleeve, and a central shaft connector is rotatably arranged in said central shaft sleeve, and said central shaft connector is rotatably connected with said movable support component; and said restorer fixing component comprises a spring first base and a spring second base which are

oppositely arranged; said elastic member comprises a restorer spring and a restoring rubber arranged between said spring first base and said spring second base; two ends of said spring second base are respectively connected with said restoring rubber and said movable central shaft block; said restorer fixing component further comprises a restorer fixing seat arranged on said fixed support component; and said restorer fixing seat is connected with said spring first base.

14. The electric skateboard according to claim 13, wherein said restorer assembly further comprises a first connecting shaft arranged in said spring first base, one end of said first connecting shaft far away from said spring first base is arranged in said movable shaft block, and a restorer adjusting nut is operatively arranged on said first connecting shaft, and said restorer adjusting nut can change an elastic force of said elastic member; and wherein, said damper assembly further comprises at least one damping adjusting device which is arranged on said damper shell and can change a damping size of said damper assembly.

15. The electric skateboard according to claim 9, wherein said moving device comprises a wheel hub and a tire fixed on said wheel hub, said front axle is arranged on said bridge arm, and said wheel hub is rotatably connected with said front axle; and said driving device comprises a driving substrate and a motor fixed on said driving substrate, wherein said motor has at least one output end, and a driving gear is arranged on said output end; and wherein, said driving device further comprises a driven gear, a transmission belt and a dust cover, wherein said driven gear is rotatably connected to said driving substrate, said transmission belt is movably connected with said driven gear and said driving gear, and said dust cover is installed on said driving substrate to cover said driving gear; and wherein said driven gear is provided with a wheel fixing groove, said wheel hub is provided with a wheel fixing plate matched with said wheel fixing groove, and said driven gear drives said wheel hub; and wherein, said driving device further comprises a motor cable clamp, and said motor cable clamp is fixed on said rear axle assembly.

16. The electric skateboard according to claim 9, wherein said electric control device comprises a control assembly, which comprises a controller shell, a light control circuit component and a first control circuit component, wherein said light control circuit component and said first control circuit component are arranged in said controller shell; said battery core is electrically connected with said light control circuit component and said first control circuit component; one end of said battery compartment cover far from said front end is connected with said control assembly; and said battery inlet circuit and said battery discharge circuit are respectively electrically connected with said control assembly

17. A buffering mechanism for an electric skateboard, comprising a support structure, a damper assembly and a restorer assembly, wherein said support structure comprises at least one fixed support component and a movable support component; said fixed support component is fixedly connected with said damper assembly and said restorer assembly; said damper assembly is provided with a damper push rod and a damper shell which can slide on said damper push rod; and said restorer assembly comprises a restorer fixing assembly and at least one elastic member; two ends of said elastic member are pressed against two ends of said restorer fixing component, said movable support component is operatively connected with said damper assembly and said restorer assembly, and said damper assembly further comprises at least one damping adjusting device which is arranged on said damper shell and can change a damping size of said damper assembly.

18. The buffering mechanism for an electric skateboard according to claim 17, wherein said movable support component comprises a bridge support, and said bridge support is provided with a central shaft cavity, a support shaft hole, a central shaft hole and a damping hole; and wherein, said fixed support component comprises a bridge arm, said bridge arm is provided with a damping push rod installation cavity and a support shaft installation cavity; said damper push rod is arranged in said damping push rod installation cavity, and a damper clamp is arranged on said damper shell; at least one sliding cavity is arranged on said damper clamp, a damper slider is arranged in said sliding cavity in a sliding way up and down, and said damper slider is rotatably connected in said

damping hole; and wherein at least one bridge damping block is fixed on said bridge support; and wherein, a support rotating shaft is rotatably arranged in said support shaft installation cavity, and said support rotating shaft is rotatably connected with said support shaft hole.

19. The buffering mechanism of said electric skateboard according to claim 17, wherein said restorer assembly comprises at least one movable central shaft block, and said movable central shaft block is provided with a central shaft sleeve, a central shaft connector is rotatably arranged in said central shaft sleeve, and said central shaft connector is rotatably connected with said central shaft hole, and said movable central shaft block is arranged in said central shaft cavity; and said restorer fixing component comprises a spring first base and a spring second base which are oppositely arranged, and said elastic member comprises a restorer spring and a restoring rubber arranged between said spring first base and said spring second base; two ends of said spring second base are respectively connected with said restoring rubber and said movable central shaft block; said restorer fixing component further comprises a restorer fixing seat arranged on said bridge arm, and said restorer fixing seat is connected with said spring first base.

20. The buffering mechanism for an electric skateboard according to claim 19, wherein said restorer assembly further comprises a first connecting shaft arranged in said spring first base, and one end of said first connecting shaft far away from said spring first base is arranged in said movable shaft block; and a restorer adjusting nut is operatively arranged on said first connecting shaft, and said restorer adjusting nut can change an elastic force of said elastic member.
